

ie-SRS

SYSTEM ADMIN PLATFORM

Comprehensive Users Guide

SONAR — Shore Outage Notification, Analysis & Response
Naval Base Facility Management & Energy Resilience Platform

For NES Practitioners, Installation Energy Managers & Base Commanders

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1. PLATFORM OVERVIEW

Mission

The ie-SRS ADMIN platform is a desktop command center for Navy Installation Energy Managers and NES practitioners. It provides unified visibility into shore facility infrastructure, energy resilience posture, smart grid operations (AMI systems), microgrid architecture, and cross-device commander communications.

Key Capabilities

- **Multi-module administration:** User accounts, messaging, alerts, audits, settings
- **Unified global map view:** Real-time installation security & resilience posture (SYS-03)
- **Data management:** Drag-and-drop data drop zone with bus-synchronized persistence (SYS-04)
- **Deployment workflows:** 14-slide deployment strategy module for site activation
- **Commander mobile bridge:** Launch mobile companion app for field operations
- **SONAR alert routing:** Configure and monitor real-time outage notifications

Platform Architecture

The ADMIN platform is part of the ie-SRS suite, which consists of:

- **ie-SRS-ADMIN.html** — This desktop admin console (Chrome-based)
- **ie-SRS.html** — Core hub for cross-app routing
- **cmdr/index.html** — Mobile commander app with SONAR composer & live alerts
- **live/cmdr.html** — Broadcast demo with animated scenario playback

All surfaces share a common event bus (*bus.js*) using GitHub Contents API and BroadcastChannel transport, with localStorage persistence for the Data Drop Zone (SYS-04).

2. KPI STRIP & KEY METRICS

Location & Purpose

The KPI Strip appears in the top chrome bar of the ADMIN platform. It displays real-time summary metrics for the selected installation:

Outages: Active shore facility outages / stalls in progress

Alerts: SONAR alerts generated and routed to commanders

Health: System uptime & resilience score (0-100)

Fleet: Connected smart meter (AMI) devices & CFE assets

Region: Current geographic region or installation selector

Status: Overall platform status (NOMINAL, ALERT, CRITICAL)

Interaction

Click on any KPI tile to drill down into the associated module:

- Outages → ADM-04 (Instance Health)
- Alerts → ADM-03 (Alert Routing)
- Health → SYS-03 (Global Map / Security & Resiliency)
- Fleet → SYS-04 (Data Drop Zone) or SYS-03 asset browser
- Region → Hub installation selector

3. ADMIN MODULES (ADM-01 THROUGH ADM-10)

ADM-01: User Accounts

Manage platform users, roles, and permissions. Add/remove users, assign role-based access controls (RBAC), configure 2FA/MFA for base commanders.

Common Tasks:

- Create new user accounts with email & role assignment
- Revoke access for departed personnel
- Toggle multi-factor authentication (MFA) enforcement
- View account audit trail & login history
- Bulk import users from directory services (LDAP/AD)

ADM-02: Message Center

Centralized messaging hub for inter-base communications. Send broadcasts, notifications, and maintenance alerts to installation commanders & energy managers.

Common Tasks:

- Compose & send platform-wide messages
- Schedule maintenance notifications
- Create distribution lists by role or region
- Track message delivery & read receipts
- Archive message history for compliance

ADM-03: SONAR Alert Routing

Configure alert thresholds, routing rules, & escalation paths. Define which events trigger SONAR notifications & which commanders receive them.

Common Tasks:

- Set outage detection thresholds (kW, duration, facility type)
- Configure alert routing by geographic region
- Define escalation (email → SMS → in-app push)
- Create alert filters (exclude test/maintenance windows)
- Test alert delivery with mock SONAR events

ADM-04: Instance Health & Deployment

Monitor platform component health. Track API uptime, database replication, bus latency, & external service connectivity. Deploy updates with zero-downtime strategy.

Common Tasks:

- View real-time component health scores
- Monitor bus message latency & throughput
- Check GitHub sync status & token health
- Review deployment history & rollback options
- Trigger staged deployments (staging → production)

ADM-05: Audit Log

Immutable record of all administrative actions. Search & export audit trails for compliance (FISMA, UFC 4-010-06), incident investigation, & forensics.

Common Tasks:

- Search audit log by user, action, timestamp, resource
- Filter by severity (INFO, WARN, ERROR, CRITICAL)
- Export audit reports in CSV/JSON for archival
- Set retention policies (30/90/365 day rolling windows)
- Alert on suspicious patterns (failed logins, privilege escalation)

ADM-06: Platform Settings

Global configuration & system preferences. Set default timeouts, data retention, UI themes, authentication backends, & GitHub organization settings.

Common Tasks:

- Configure GitHub org, repo, & branch for deployment
- Set session timeout & inactivity auto-logout
- Choose UI theme (dark/light) & language preferences
- Enable/disable experimental features by role
- Configure external service integrations (Slack, webhooks)

ADM-07: Deployment Strategy

14-slide guided workflow for site activation. Step-by-step playbook for deploying ie-SRS to new naval bases, including pre-flight checks, data migration, & go-live signoff.

Common Tasks:

- Review pre-deployment readiness checklist
- Import baseline installation data (ADMIN_CORE dataset)
- Configure per-site AMI/FRCS/microgrid parameters
- Run site connectivity & API endpoint validation
- Execute & monitor data migration from legacy systems

ADM-08: SONAR — Shore Outage Notification, Analysis & Response

Real-time outage detection & notification engine. Monitors AMI smart meters & FRCS control systems for supply interruptions & analyzes impact on microgrid stability.

Common Tasks:

- View active outages with timeline & affected assets
- Analyze outage cause (weather, equipment failure, scheduled)
- Compare impact across multiple installations
- Manually resolve stalled outages with analyst override
- Export incident reports for post-action reviews (AARs)

ADM-09: Comms Log

Complete record of all platform communications. SONAR alerts, commander acknowledgments, message center broadcasts, & mobile-to-mobile chats logged here.

Common Tasks:

- Search comms by sender, recipient, alert ID, date range
- Filter by message type (WARNORD, DROP, acknowledgment, status update)
- View acknowledgment times & response metrics
- Export comms transcripts for after-action reviews
- Correlate alerts with commander responses

ADM-10: CMDR Mobile

Launch & configure the mobile commander companion app. See linked mobile commanders, push test alerts, & monitor real-time acknowledgments from the field.

Common Tasks:

- View active CMDR mobile sessions (linked commanders)
- Generate & display QR code for mobile app pairing
- Push test SONAR alerts to mobile devices
- Monitor mobile-to-mobile message routing
- Configure transport (same-browser or cross-device GitHub)

4. SYSTEM MODULES (SYS-03, SYS-04)

SYS-03: Global Map & Security/Resiliency

Location: Hub card labeled "Security & Resiliency · Global Map"

Purpose: Real-time interactive map of all Naval Base installations with energy resilience posture visualization. Shows FRCS cybersecurity exposure, microgrid readiness, CFE asset health, & geographic storm/threat zones.

Key Features:

- Leaflet map with ArcGIS World Imagery underlay
- San Diego regional coordinates (NASNI Coronado, MCAS Miramar, Harbor Drive corridor)
- Interactive installation markers with drill-down detail panels
- Real-time CFE asset data overlay (14,968 assets in DUERS_LOCKED dataset)
- Threat/storm zone heatmaps
- Convoy & logistics route visualization (I-15, Coronado Bridge)

Common Tasks:

- Click on installation marker → see facility details, asset counts, resilience score
- Hover over asset cluster → see aggregated health metrics
- Toggle overlays (FRCS zones, microgrid boundaries, storm tracks)
- Export map snapshot for briefings

SYS-04: Data Drop Zone

Location: Hub card labeled "Data Drop Zone"

Purpose: Universal drag-and-drop import/export interface for site data. Supports CSV, JSON, & Excel ingestion. Syncs across all ie-SRS surfaces via the shared event bus with localStorage persistence.

Key Features:

- Drag-and-drop file upload (CSV, JSON, XLSX, GeoJSON)
- Real-time preview & validation of imported data
- Automatic schema mapping (installation, asset, alert, metric records)
- Bus-synchronized updates across all open windows/tabs
- localStorage-backed persistence (survives refresh)
- Export current session data as CSV/JSON download

Common Tasks:

- Drag CSV file → preview mapped columns → confirm import
- Check sync status indicator (green = synced, yellow = pending)
- Clear stale records with delete controls
- Export working dataset for external analysis

5. COMMON WORKFLOWS

Workflow 1: Respond to a Shore Outage

Trigger: Red alert notification in ADM-03 (SONAR Alert Routing) or Comms Log (ADM-09)

1. **Navigate to ADM-04 (Instance Health).** See active outages list with duration, facility, & affected asset count.
2. **Click on the outage** to open detail panel. Review timeline, suspected cause, & impacted circuits.
3. **Cross-reference SYS-03 (Global Map).** Zoom to facility location. Check for weather/threat overlay match.
4. **Notify field teams via ADM-10 (CMDR Mobile).** Push alert to mobile commanders. Monitor acknowledgment in Comms Log.
5. **Update status in ADM-04** as repair progresses (Investigating → In Progress → Resolved).
6. **Export incident report (ADM-05 Audit Log, ADM-09 Comms Log)** for after-action review.

Workflow 2: Onboard a New Installation

Trigger: New Naval Base site activation request

1. **Open ADM-07 (Deployment Strategy).** Step through the 14-slide deployment checklist.
2. **Pre-flight:** Verify site name, coordinates, facility type, AMI device count, FRCS zone assignments.
3. **Data Migration:** Use SYS-04 (Data Drop Zone) to import baseline asset CSV from ADMIN_CORE dataset.
4. **Map Configuration:** In SYS-03 (Global Map), set site marker coordinates & resilience baseline.
5. **Create admin user:** Return to ADM-01 (User Accounts). Create base commander & energy manager accounts with RBAC.
6. **Configure alerts:** In ADM-03 (Alert Routing), set outage thresholds & routing rules for the new site.
7. **Test deployment:** In ADM-10 (CMDR Mobile), pair a test mobile device & send mock SONAR alert.
8. **Go-live:** Final sign-off in ADM-04 (Instance Health). Update site status to OPERATIONAL.

Workflow 3: Investigate a Security Anomaly

Trigger: Suspicious pattern detected in ADM-05 (Audit Log) or FRCS alert in SYS-03

1. **Open ADM-05 (Audit Log).** Search by timeframe, user, or action type. Filter for privilege escalation, failed logins, config changes.
2. **Correlate with Comms Log (ADM-09).** Look for messaging or alert activity during same window.
3. **Cross-check SYS-03 (Global Map).** Check FRCS cybersecurity zones for any exposure indicated near suspect activity.
4. **Escalate if needed:** Use ADM-02 (Message Center) to broadcast security alert to all commanders.
5. **Isolate & remediate:** Revoke suspect user account (ADM-01). Force password reset for affected users.
6. **Document:** Export full audit trail for incident response team & DoD security authorities.

6. TROUBLESHOOTING & BEST PRACTICES

Best Practices

- **Monitor KPI Strip regularly.** The KPI strip is your at-a-glance health dashboard. Check it every 2 hours during operational windows.
- **Use SYS-03 & SYS-04 for data confidence.** Always cross-reference the global map with drop-zone data before making decisions.
- **Enable MFA for all commanders.** Enforce 2FA/MFA in ADM-01 for users with outage response authority.
- **Audit weekly.** Schedule weekly ADM-05 audit log reviews to catch anomalies early.
- **Test alert routing monthly.** Use ADM-03 to run mock SONAR exercises & verify commander receipt.
- **Archive compliance logs.** Export ADM-05 & ADM-09 monthly for FISMA/UFC compliance archival.
- **Keep GitHub token fresh.** Check ADM-04 (Instance Health) for GitHub API token expiry. Rotate tokens quarterly.
- **Document configuration changes.** Every change in ADM-06, ADM-03, or ADM-07 should be logged with a reason in message center (ADM-02).

Common Issues & Resolutions

Data Drop Zone not syncing: Check bus status in ADM-04. Reload page & verify localStorage is enabled in browser. Check GitHub token health.

SONAR alerts not routing to CMDR mobile: Verify ADM-10 device is paired & transport mode set correctly (same-browser vs. cross-device). Check ADM-03 alert routing rules.

SYS-03 map not loading installation markers: Zoom to San Diego region (coordinates ~32.7°N, 117.2°W). Check ArcGIS imagery layer connectivity. Clear browser cache.

ADM-07 deployment stuck on validation: Review pre-flight checklist in slide 2. Ensure all required fields filled: site name, lat/lon, facility type, CFE asset count.

Audit log not showing recent actions: Wait 30 seconds for log flush. Refresh ADM-05 page. Check your user role has audit read permission in ADM-01.

APPENDIX: KEYBOARD SHORTCUTS & REFERENCE

Navigation

Key	Action
M	Open Message Center (ADM-02)
U	Open User Accounts (ADM-01)
A	Open Alert Routing (ADM-03)
H	Open Instance Health (ADM-04)
L	Open Audit Log (ADM-05)
S	Open Settings (ADM-06)
D	Open Deployment Strategy (ADM-07)
C	Open CMDR Mobile (ADM-10)
G	Open Global Map (SYS-03)
Z	Open Data Drop Zone (SYS-04)
? or F1	Open help & shortcuts

Key Terminology

- **AMI:** Advanced Metering Infrastructure (smart grid meters)
- **FRCS:** Facility Remote Control System (SCADA cybersecurity framework)
- **CFE:** Common Fleet Elements (Navy standard equipment taxonomy)
- **NES:** Navy Energy Security initiative
- **NAVFAC:** Naval Facilities Engineering Command
- **RMF/ATO:** Risk Management Framework / Authority to Operate (DoD compliance)
- **SONAR:** Shore Outage Notification, Analysis & Response
- **BLACKSTART:** Rapid facility resilience restart protocol

Support & Resources

Platform Documentation: <https://thompsonryane-collab.github.io/ie-SRS/>

Report Issues: Contact your NAVFAC NES technical lead or submit a bug report via the Help icon (?) in the ADMIN platform chrome bar.

Admin Support: See ADM-01 for contact information for your installation's designated platform administrator.

Version: ie-SRS v1.0 (SONAR • BLACKSTART • Grid Resilience)

This guide reflects ie-SRS ADMIN platform as of {datetime.now().strftime('%B %Y')}. Features and workflows subject to change with platform updates.